

Ayurveda IPR and Regulatory Assistant — Full Technical Proposal

Comprehensive Research-Backed Implementation Blueprint

Generated: August 26, 2026 | Based on deep research of PS requirements

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1. Executive Summary

- This proposal outlines the **complete technical blueprint** for an **Ayurveda IPR and Regulatory Assistant** — a domain-expert legal AI system that combines:
- **6 core capabilities:** Formulation Classification, IPR Advisory, ABS Compliance, TKDL/Prior Art Search, Jurisdiction Toggle (India/International), and Source-Cited Legal Answers
 - **Multilingual voice-first interface** in Hindi (Devanagari), Tamil, and English
 - **Cross-platform delivery** via React Native (Web + iOS + Android)
 - **Open-source AI stack** with RAG, knowledge graph, OCR, and speech

IMPORTANT

****No existing tool combines all 6 capabilities.**** This is a genuine market gap. IP SAARTHI, KanoonGPT, Formulaite each cover only 1-2 aspects.

2. Data Scraping — Exact Sources, Methods and Pipelines

2.1 Complete Data Source Matrix (23 Sources)

Layer 1: Statutes and Rules (The Law Itself)

#	Source	URL	What to Extract	Scraping Method	Format	Priority
1	India Code	indiacode.nic.in	17+ Central Acts (Patents Act 1970, TM Act 1999, GI Act 1999, Biodiversity Act 2002, D&C Act 1940)	requests + BeautifulSoup; community Statute API on GitHub for JSON	HTML to JSON	P0
2	Gazette of India	egazette.gov.in	Patent Rules 2024, Biodiversity Rules 2024, ABS Regulations 2025, D&C Rules 1945 amendments	PDF download + PyMuPDF text extraction	PDF to Text	P0
3	AYUSH Patent Guidelines 2025	ipindia.gov.in	AYUSH-specific patent examination guidelines	Direct PDF download	PDF to Text	P0
4	WIPO Lex	wipo.int/wipolex	TRIPS, CBD, Nagoya Protocol, WIPO GRATK Treaty 2024, PCT, Madrid, Hague, Budapest	Partial REST API; bulk PDF download	PDF/HTML to Text	P0
5	CBD ABSCH	absch.cbd.int	ABS permits, Nagoya checkpoint data	REST API (JSON responses)	JSON	P1

Layer 2: Pharmacopoeial and Classical Texts

#	Source	URL	What to Extract	Scraping Method	Format	Priority
6	Ayurvedic Pharmacopoeia of India	archive.org, pcimh.gov.in	Drug monographs: Name, Ingredients, Method, Standards	Download PDFs, OCR with Surya OCR for scanned pages	PDF (scanned) to OCR to JSON	P0
7	Ayurvedic Formulary of India	archive.org, pcimh.gov.in	Classical formulations in First Schedule texts	Same as above	PDF to JSON	P0
8	e-Samhita	niimh.nic.in/ebooks	Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya	Web scraping with structured chapter/verse extraction	HTML to JSON	P1
9	DRAVYA Portal (CCRAS)	ccras.nic.in	Ayurvedic substance database — botanical names, parts used	Playwright (dynamic JS pages)	HTML to JSON	P1
10	FSSAI Ayurveda Aahar	fssai.gov.in	Schedule A — Category A product recipes	PDF extraction	PDF to JSON	P1

Layer 3: Registry and Case Data

#	Source	URL	What to Extract	Scraping Method	Format	Priority
11	InPASS (Patent Search)	ipindiaservices.gov.in/publicsearch	Indian AYUSH patent applications and grants	Playwright (no API)	HTML to JSON	P1
12	Google Patents	patents.google.com	Global patents including India — bulk AYUSH data	Google Patents BigQuery (official, free tier)	JSON	P0
13	WIPO PATENTSCOPE	patentscope.wipo.int	PCT applications for herbal/Ayurvedic inventions	REST API (official)	XML/JSON	P1
14	Espacenet	worldwide.espacenet.com	EPO patent database — global herbal patents	Open Patent Services (OPS) API	XML	P2
15	TMR India (Trademark)	ipindiaservices.gov.in/tmrpublicsearch	Ayurvedic brand name trademarks	Playwright	HTML to JSON	P2
16	GI Registry	ipindia.gov.in/geographical-indications.htm	Registered GIs for Ayurvedic products	PDF list downloads	PDF to JSON	P2
17	IndianKanoon	indiankanoon.org	IPR case law, Ayurveda judgments	HuggingFace dataset: d-riti/Dataset-For-Indian-Legal-Knowledge-Base	JSON/Parquet	P1
18	eCourts India	ecourts.gov.in	Official case records, orders	eCourtsIndia API (authorized)	JSON	P2
19	NBA ABS Portal	absefiling.nbaindia.in	ABS certificates, filings	Web scraping	HTML to JSON	P2

Layer 4: TKDL (Special Handling)

#	Source	Status	Strategy
20	TKDL (tkdl.res.in)	RESTRICTED — 500K+ formulations; paid subscription	Cannot directly ingest. Build "TKDL Pointer": reference TKRC codes in KB, guide users to search TKDL, use as prior art reference in IPR routing

Layer 5: Pre-Built Datasets

#	Source	Contents	Access
21	KanoonGPT Dataset (HuggingFace)	Cleaned Indian legal text	HuggingFace API
22	Indian Legal Corpus (HuggingFace)	Structured Indian legal docs	HuggingFace API
23	Ayusoft	Ayurvedic knowledge repository	Web access

2.2 Data Ingestion Pipeline

COLLECTORS

Web Scrapers (Playwright, BS4, requests)

API Clients (WIPO, BigQuery)

HF Datasets

-->

PROCESSORS

PDF Parser (PyMuPDF)

OCR (Surya OCR)

HTML Parser (BS4)

Legal Section Splitter (custom hierarchical)

-->

STRUCTURED OUTPUT

Section-Level Chunks with:

statute_name, section_number,

jurisdiction, last_amended_date,

cross_references[], chunk_hash

Git-Versioned Corpus (DVC)

2.3 Chunking Strategy (CRITICAL)

IMPORTANT

NEVER use naive fixed-token splitting on legal text. It destroys section boundaries and makes citations impossible.

Document Type	Chunking Unit	Max Chunk	Overlap	Key Metadata
Statutes	Section / Sub-section	1500 tokens	200	act_name, section_number, year
Case Law	Paragraph	2000 tokens	300	case_citation, court, year
Pharmacopoeia	Drug Monograph	1000 tokens	100	drug_name, ingredients[]
Treaties	Article	1500 tokens	200	treaty_name, article_number

2.4 Scraping Toolkit

Tool	Use Case	Why
Playwright (Python)	Dynamic JS portals (InPASS, TMR, DRAVYA)	CAPTCHA, JS rendering, pagination
requests + BeautifulSoup	Static HTML (India Code, WIPO Lex)	Lightweight, fast
PyMuPDF (fitz)	PDF extraction (Gazette, rules, treaties)	Fast, complex layouts
Surya OCR	Scanned PDFs (old Pharmacopoeia)	90+ languages, layout-aware
Google BigQuery	Bulk patent data	Free tier sufficient
HuggingFace datasets	Legal corpora (ILC, KanoonGPT)	One-line download

3. Cross-Platform App — React Native (Web + iOS + Android)

3.1 Technology Stack

One codebase for Web + iOS + Android (like Claude app). Gold standard in 2026: **Expo SDK 55+ with New Architecture**.

Component	Technology	Notes
Framework	React Native + Expo SDK 55+	New Architecture (Fabric + TurboModules)
Routing	Expo Router (file-based)	Shared navigation, deep linking
Runtime	Hermes engine	AOT compilation, fast cold starts
Language	TypeScript (strict)	Non-negotiable
State (Local)	Zustand	Lightweight
State (Server)	TanStack Query v5	Caching, background sync
Styling	NativeWind (Tailwind for RN)	Shared design system
Audio Capture	expo-audio + @mykin-ai/expo-audio-stream	Real-time PCM streaming
Audio Playback	expo-av	Streamed TTS
WebSocket	Native WebSocket API	Binary PCM streaming
Camera/OCR	expo-camera + expo-image-picker	Document scanning
Build/CI	EAS (Expo Application Services)	Cloud builds, OTA
Animations	react-native-reanimated	60/120 FPS

3.2 Project Structure

```
ayurveda-ipr-app/  
app/                                # File-based routes (Expo Router)  
  (tabs)/  
    chat.tsx                        # Main chat interface  
    scan.tsx                       # OCR document scanner  
    classify.tsx                   # Formulation classifier wizard  
    settings.tsx                 # Language, jurisdiction, profile  
    _layout.tsx                  # Root layout  
    index.tsx                    # Landing/onboarding  
components/  
  chat/  
    MessageBubble.tsx            # Chat message with citation links  
    CitationCard.tsx            # Expandable citation reference  
    ConfidenceIndicator.tsx      # Traffic light (High/Med/Low)  
    JurisdictionToggle.tsx      # [India | International] switch  
    VoiceInput.tsx              # Mic button with VAD + waveform  
    EscalationButton.tsx        # Connect to IP facilitator  
  classifier/  
    ClassificationWizard.tsx     # 5-7 step decision tree  
    ResultCard.tsx              # Result + reasoning  
  scanner/  
    DocumentScanner.tsx         # Camera-based capture  
    OCRResultView.tsx           # Extracted text with edit  
  shared/  
    DisclaimerBanner.tsx        # Not legal advice  
    LanguageSelector.tsx        # Hindi / Tamil / English  
services/  
  api/ (chatService, ocrService, voiceService, classifierService)  
  auth/ (authService - DPDP-compliant)  
  storage/ (chatHistory - local encrypted)  
hooks/ (useVoiceRecording, useChat, useJurisdiction, useLanguage)  
constants/ (theme, languages, config)  
types/ (chat, classification, api)
```

3.3 Voice Chat Flow (Low Latency)

```
User speaks Hindi  
|  
expo-audio-stream (16kHz PCM, 100ms buffers)  
|  
WebSocket (Binary, raw PCM - NO Base64)  
|  
BACKEND:  
  PCM --> IndicConformer ASR --> Hindi Text  
    --> NMT (optional) --> English Text  
    --> RAG Pipeline --> English Answer  
    --> NMT (optional) --> Hindi Text  
    --> IndicParler TTS --> Hindi Audio PCM chunks  
|  
WebSocket (Binary) --> expo-av playback  
|  
User hears response while LLM still generating
```

3.4 Key UI Features

Feature	Implementation
Chat Interface	Scrollable messages + input bar + mic (like Claude)
Jurisdiction Toggle	Segmented [India
Citation Links	Inline [Section 3(p), Patents Act 1970] tappable links
Confidence	Traffic light: Green High / Yellow Medium / Red Low
Disclaimer	Persistent "This is information, not legal advice"
Escalation	"Connect to IP Facilitator" button
Document Scanner	Camera capture, auto-crop, OCR, extract details
Language	Three-way: English / Hindi / Tamil

4. AI LLM — Multilingual Legal Reasoning with RAG

4.1 Core Requirements

- 1. Reason about legal text, interpret statutes, cite sources
- 2. Understand Hindi (Devanagari script) and Tamil natively
- 3. Support RAG with grounding, never hallucinate
- 4. Open-source, self-hostable for DPDP compliance

4.2 Model Comparison

Model	Params	Active	Context	Hindi/Tamil	Legal	License	GPU
Sarvam-105B	105B	10B MoE	65K	5/5 native 22 langs	4/5	Apache 2.0	4x A100
Qwen3-235B	235B	22B MoE	128K	4/5 strong multilingual	5/5 dual thinking	Apache 2.0	8x A100
Qwen3-30B-A3B	30B	3B MoE	128K	4/5	4/5	Apache 2.0	1x A100
Sarvam-30B	32B	2.4B MoE	65K	5/5	3/5	Apache 2.0	1x A100
DeepSeek-V4-Pro	Large	37B	128K+	3/5	5/5	Open	6x A100

4.3 Recommended: Dual-Model Strategy

TIP

Use TWO models. This pattern wins hackathons and production deployments.

MODEL 1: Sarvam-105B (Sarvam-30B for MVP) = "The Indian Language Expert"

- Native Hindi/Tamil in Devanagari/Tamil script
- User-facing chat, cultural context, AYUSH terminology

MODEL 2: Qwen3-30B-A3B (Qwen3-235B for production) = "The Legal Reasoning Engine"

- Dual-mode thinking for complex IPR analysis
- 128K context for long statutes
- Citation extraction, classification, IPR routing

ROUTER: LangGraph Orchestrator decides which model handles each part.

4.4 Embedding Models

Model	Purpose	Why
BGE-M3	Primary embeddings	Hybrid dense+sparse; multilingual; legal RAG standard
IndicBERT v2 / MuRIL	Hindi/Tamil specific	Custom retrieval over pharmacopoeial texts

4.5 RAG System Prompt

You are an Ayurveda IPR and Regulatory Assistant.

RULES:

1. ONLY use information from provided context documents.
2. EVERY claim MUST cite: [Section X, Act Name, Year]
3. If insufficient info: say so and recommend IP attorney. [Escalate]
4. Append: "This is information, not legal advice."
5. Confidence: HIGH (exact match) | MEDIUM (related) | LOW (limited)
6. Jurisdiction: INDIA = Indian statutes | INTERNATIONAL = Treaties
Keep both visually separated.

CONTEXT: {retrieved_context}

QUERY (Jurisdiction: {jurisdiction}): {user_query}

5. Speaking Model — TTS for Indian Languages

5.1 Model Comparison

Model	Languages	Quality	Latency	License
IndicParler-TTS (AI4Bharat)	18-23 Indian	SOTA 5/5	500ms	Apache 2.0
svara-TTS	19 Indian	Emotional 5/5	600ms	Open
Sooktam2 (BharatGen)	Hindi, Tamil, etc.	4/5	400ms	Open
XTTS v2 (Coqui)	17+ multilingual	4/5	700ms	Apache 2.0
Bhashini TTS	22 Indian	3/5	800ms	Free (govt)
Kokoro (82M)	Limited Indic	3/5	100ms ultra-fast	Open

5.2 Recommended Strategy

- **PRIMARY:** IndicParler-TTS — best Hindi/Tamil quality, Apache 2.0, self-hosted
- **FALLBACK:** Bhashini TTS API — free government API, 22 languages

- **EDGE:** Kokoro (82M) — CPU-only, for short confirmations

5.3 Streaming Architecture

```
LLM generates tokens (streaming)
--> Buffer to sentence boundary
--> IndicParler-TTS generates audio chunk
--> WebSocket streams PCM to client
--> expo-av real-time playback
= User hears while LLM still generating
```

6. Speech Recognition — ASR for Indic Languages

6.1 Model Comparison

Model	Architecture	Languages	WER Hindi	Code-Mix	Latency	License
IndicConformer (AI4Bharat)	Conformer CTC-RNNT	22 Indian	~8%	5/5	<100ms streaming	MIT
Shunya/Pingala V1	Fine-tuned Whisper	Indian	~10%	5/5 best code-switch	~300ms	Open
faster-whisper	Whisper+CTranslate2	Multilingual	~12%	3/5	~200ms	MIT
Saaras V3 (Sarvam)	Custom	Indian	~9%	5/5	<100ms	Open
Bhashini ASR	ULCA models	22 Indian	~15%	3/5	~500ms	Free

6.2 Recommended Strategy

- **PRIMARY:** IndicConformer — gold standard, 600M params, streaming, VEXYL-STT Docker
- **SECONDARY:** Shunya/Pingala — best code-mixed Hindi+English, Tamil+English
- **FALLBACK:** Bhashini ASR API — free, zero cost, good for MVP

6.3 Custom Legal Vocabulary

WARNING

Standard ASR struggles with legal terms. Add hotword boosting for: Section 3(p/d/e), Patents Act, Trademarks Act, GI Act, Biodiversity Act, FSSAI, TKDL, ABS, Nagoya Protocol, WIPO GRATK, and Hindi/Tamil equivalents.

7. OCR and Document Understanding

7.1 Two Use Cases

1. **Corpus Building:** OCR scanned Pharmacopoeia (batch)
2. **Real-Time:** User photographs document, extract and understand within seconds

7.2 Model Comparison

Model	Use Case	Languages	Speed	Layout	License
Surya OCR 2	Batch + Real-time	90+ (Hindi, Tamil)	4/5	5/5	GPL-3.0
PaddleOCR	High-throughput	Many	5/5 fastest	5/5	Apache 2.0
Qwen2.5-VL 7B	Document reasoning	Universal	3/5 heavier	5/5 OCR-free	Apache 2.0
InternVL3	Document reasoning	Universal	3/5	5/5	Open

7.3 Two-Tier Strategy

TIER 1: Surya OCR 2 (Fast Text Extraction)

- Text detection + recognition in one pass
- Hindi Devanagari + Tamil native support
- ~200ms/page (GPU), ~1s/page (CPU)
- Output: Structured Markdown

TIER 2: Qwen2.5-VL 7B (Deep Document Understanding)

- OCR-free document reasoning
- Answers: "What ingredients?", "Classical or proprietary?"
- ~2-5s per document (GPU)

FLOW: User photo --> Surya OCR (fast text, display) --> Qwen2.5-VL (deep understanding) --> Auto-populate classification wizard

7.4 Performance Targets

Metric	Target	How
OCR text extraction	< 500ms/page	Surya on GPU
Document understanding	< 3 seconds	Qwen2.5-VL 7B + vLLM
Batch processing	100 pages/min	PaddleOCR GPU batching
Mobile photo to text	< 2s end-to-end	Client preprocess + server

8. RAG Database Architecture

8.1 Why Hybrid RAG

Legal RAG needs multi-hop reasoning. Example: "Can I patent my Ashwagandha formulation?" touches:

1. Section 3(p) — traditional knowledge bar
2. AYUSH Guidelines 2025 — inventive step test
3. TKDL prior art — documented formulations
4. Biodiversity Act 2002 — ABS requirements
5. Section 3(e) — mere admixture bar

Simple vector search cannot traverse these. Need **Vector + Graph + BM25**.

8.2 Architecture Flow

```
USER QUERY
|
Intent Parser --> Query Decomposer (sub-queries)
|
HYBRID RETRIEVAL
  Vector Search (Qdrant + BGE-M3) + BM25 (Elasticsearch)
  |
  Reciprocal Rank Fusion (RRF, k=60)
  |
  Cross-Encoder Reranker (bge-reranker-v2-m3)
  |
  KNOWLEDGE GRAPH (Neo4j)
  Traverse: cross-refs, amendments, interpretations, product routing
  Output: Annotated subgraph with provenance
  |
CONTEXT ASSEMBLY --> LLM --> Cited Answer + Confidence Score
```

8.3 Technology Choices

Component	Technology	Why
Vector Store	Qdrant (self-hosted)	Speed, hybrid search, payload filtering
Keyword Search	Elasticsearch (rank_bm25 for MVP)	Exact section number matching
Knowledge Graph	Neo4j Community	Cypher queries, native vector since v5.x
Embeddings	BGE-M3	Dense+sparse+ColBERT; multilingual
Reranker	bge-reranker-v2-m3	Highest-ROI for legal precision
Cache	Redis	Response caching, sessions
Blob Storage	MinIO (S3-compatible)	PDFs, OCR results, audit

8.4 Neo4j Knowledge Graph Schema

Node Types: Statute, Section, SubSection, Rule, Treaty, Article, CaseLaw, DrugFormulation, Ingredient, DrugMonograph, IPCategory, Jurisdiction, PatentApplication, TKDLRecord, ABSRequirement, RegulatoryPathway

Key Relationships:

- Section BELONGS_TO Statute
- Section REFERS_TO Section (cross-references)
- Section AMENDED_BY Rule
- Section INTERPRETED_IN CaseLaw
- Treaty IMPLEMENTED_BY Statute (e.g., TRIPS to Patents Act)
- DrugFormulation CLASSIFIED_AS Type
- DrugFormulation CONTAINS Ingredient
- Ingredient REQUIRES_ABS_IF ABSRequirement
- Ingredient FOUND_IN_TKDL TKDLRecord
- PatentApplication SUBJECT_TO Section

8.5 Data Sizing

Layer	Volume	Storage
Statutes and Rules	~5K chunks	~50 MB
Case Law (IPR)	~20K chunks	~200 MB
Pharmacopoeia	~8K chunks	~80 MB
Treaties	~3K chunks	~30 MB
Patent Data	~50K records	~500 MB
Knowledge Graph	~100K nodes, ~500K edges	~2 GB
Total	~86K chunks	~3 GB

9. Entire App Architecture — End-to-End

9.1 Six-Layer System Design

```
LAYER 1: CLIENT (React Native + Expo)
  Web / iOS / Android --> Expo Router (shared codebase)
    | HTTPS / WSS

LAYER 2: API GATEWAY (Nginx/Caddy)
  SSL termination, rate limiting, WebSocket upgrade
    |

LAYER 3: BACKEND (FastAPI)
  /api/chat, /api/classify, /api/ocr, /ws/voice, /api/escalate, /api/audit
    |

LAYER 4: ORCHESTRATION (LangGraph Multi-Agent)
  Intent Parser --> Query Decomposer --> Specialist Agents:
    Formulation Classifier Agent
    IPR Router Agent
    ABS Compliance Agent
    TKDL/Prior Art Agent
    Citation Grounding Agent
    |

LAYER 5: AI MODEL SERVING
  LLM Server      ASR Server      TTS Server      OCR/VLM Server
  vLLM serving    IndicConformer  IndicParler-TTS  Surya + Qwen2.5-VL
  Sarvam + Qwen3  :8001           :8002           :8003
  :8000
  |

LAYER 6: DATA and STORAGE
  Qdrant      Neo4j      Elasticsearch  Redis
  :6333       :7474     :9200          :6379
  MinIO       Postgres   Git + DVC
  :9000       :5432     (versioned corpus)
```

9.2 Service Communication

From to To	Protocol	Purpose
Client to Gateway	HTTPS/WSS	All user interactions
FastAPI to LangGraph	Python async	Agent orchestration
LangGraph to vLLM	HTTP (OpenAI-compatible)	LLM inference
LangGraph to Qdrant	gRPC/REST	Vector search
LangGraph to Neo4j	Bolt (Cypher)	Graph traversal
LangGraph to Elasticsearch	REST	BM25 keyword search
FastAPI to ASR/TTS	HTTP/WebSocket	Speech I/O
FastAPI to OCR	HTTP	Document processing
FastAPI to Redis	Redis protocol	Caching, sessions
FastAPI to Postgres	asyncpg	Users, audit logs

9.3 Docker Compose Deployment

All services containerized. Key GPU reservations:

- LLM Sarvam-105B: 4x NVIDIA GPU via vLLM
- LLM Qwen3-30B: 1x NVIDIA GPU via vLLM
- ASR/TTS/OCR: 1x NVIDIA GPU each
- Data stores: Standard containers

10. Every Other Key Necessity

10.1 Security and DPDP Compliance

Requirement	Implementation
Data Consent	DPDP Act 2023 explicit opt-in
Data Minimization	Auto-delete configurable period
Encryption at Rest	AES-256 (pgcrypto, MinIO SSE)
Encryption in Transit	TLS 1.3 (Nginx)
Audit Trail	Full query/response log in PostgreSQL
Right to Erasure	DELETE /api/user/data endpoint
Access Control	JWT + refresh; RBAC (User, Admin, Facilitator)
Rate Limiting	Redis token bucket per user

10.2 Evaluation and QA

Framework	Measures
RAGAS	Faithfulness, relevancy, context precision/recall
DeepEval	Hallucination detection, toxicity, bias
Custom Legal Eval	Citation correctness

10.3 Confidence Scoring Algorithm

```
confidence = 0.4 * top_retrieval_score
            + 0.3 * min(graph_path_depth / 3, 1.0)
            + 0.3 * min(citation_count / 3, 1.0)

if confidence >= 0.7 and citations >= 2: HIGH (green)
elif confidence >= 0.4 and citations >= 1: MEDIUM (yellow)
else: LOW (red) – recommend IP attorney
```

10.4 Escalation Triggers

Auto-escalate when: confidence < 0.3, Section 3(e) admixture analysis, TKDL access needed, international filing strategy, or user requests human expert.

10.5 Multilingual Legal Glossary

English	Hindi (Devanagari)	Tamil
Patent	पेटेंट	காப்புரிமை
Trademark	व्यापार चिह्न	வணிகக் குறி
Geographical Indication	भौगोलिक संकेत	புவிச்சான்று
Prior Art	पूर्व कला	முன்னோடி கலை
Traditional Knowledge	पारंपरिक ज्ञान	பாரம்பரிய அறிவு
Formulation	सूत्रीकरण	சூத்திரம்
Classical Medicine	शास्त्रीय औषधि	பாரம்பரிய மருந்து

10.6 Bhashini Integration

- Register at bhashini.gov.in, get ULCA API credentials
- Pipeline: ASR (hi-conformer) --> NMT (hi to en) --> RAG --> NMT (en to hi) --> TTS
- Chained ASR+NMT in single API call for reduced latency
- Use Bhashini for ASR/NMT (free), self-hosted IndicParler for TTS (better quality)

10.7 Corpus Versioning

```
corpus/
.git/           # Git versioning
.dvc/          # DVC for large files
statutes/
  patents_act_1970/
  biodiversity_act_2002/
  dc_act_1940/
treaties/
  trips/
  nagoya_protocol/
  wipo_gratk_2024/
pharmacopoeia/
  api_vol1/
  api_vol2/
case_law/
  ipr_judgments/
CHANGELOG.md
VERSION
```

10.8 Monitoring

Tool	Purpose
Prometheus + Grafana	System metrics (GPU, latency)
LangSmith / LangFuse	LLM tracing (tokens, prompts)
Sentry	Error tracking
Custom Dashboard	RAGAS scores, citation accuracy

11. Cost Estimation

Full Production

Component	Spec	Monthly INR
GPU (LLM)	4x A100 80GB	2-3 Lakh
GPU (ASR/TTS/OCR)	2x A100 40GB	80K-1.5L
CPU Server	32 vCPU, 128GB RAM	30-50K
Bhashini API	Free	0
Total		3.1-5 Lakh/month

MVP Optimized

Component	Spec	Monthly INR
GPU	1x A100 80GB (all models)	70K-1L

Component	Spec	Monthly INR
CPU Server	16 vCPU, 64GB RAM	15-25K
Total		85K-1.25L/month

Hackathon/Demo

Component	How	Monthly INR
LLM	Groq/Together API	~5K
ASR/TTS	Bhashini (free)	0
OCR	Surya local/Colab	0
Hosting	Render/Railway	2K
Total		7-15K/month

12. Development Roadmap (12 Weeks)

Phase 0: Foundation (Week 1)

- ☐ Monorepo setup (Expo + FastAPI + scrapers)
- ☐ Register on Bhashini
- ☐ Docker containers (Qdrant, Neo4j, Elasticsearch)
- ☐ Begin corpus building

Phase 1: Data Pipeline (Weeks 2-3)

- ☐ Scrape 15 core Indian statutes
- ☐ OCR Pharmacopoeia with Surya
- ☐ Structure WIPO treaties
- ☐ Hierarchical chunking to Qdrant + Elasticsearch
- ☐ Begin Neo4j graph population
- ☐ Git + DVC corpus versioning

Phase 2: RAG Core (Weeks 4-5)

- ☐ Deploy Qwen3-30B via vLLM
- ☐ Hybrid retrieval (vector + BM25 + RRF + reranker)
- ☐ Citation enforcement prompt
- ☐ Confidence scoring
- ☐ Formulation classifier agent
- ☐ IPR router + ABS compliance agents
- ☐ RAGAS evaluation pipeline

Phase 3: React Native App (Weeks 6-7)

- [] Expo SDK 55 + TypeScript project
- [] Chat UI (messages, citations, confidence)
- [] Jurisdiction toggle + classification wizard
- [] Voice I/O (Bhashini ASR + IndicParler TTS)
- [] Document scanner + OCR
- [] Language selector + escalation

Phase 4: Knowledge Graph + Multi-Agent (Weeks 8-9)

- [] Complete Neo4j population
- [] GraphRAG (vector + graph traversal)
- [] TKDL pointer module
- [] Case law integration
- [] LangGraph multi-agent orchestration

Phase 5: Polish and Production (Weeks 10-12)

- [] Deploy Sarvam-105B for production Hindi/Tamil
- [] DPDP compliance (consent, audit, encryption)
- [] Performance optimization (caching, quantization)
- [] User testing with AYUSH practitioners
- [] RAGAS benchmarks (target: >0.85 faithfulness)
- [] EAS builds (iOS + Android) + web deploy

Quick Reference

Model Specifications

Component	Model	Params	Active	Context	License	GPU
LLM (Hindi)	Sarvam-105B	105B	~10B MoE	65K	Apache 2.0	4xA100
LLM (Legal)	Qwen3-30B-A3B	30B	3B MoE	128K	Apache 2.0	1xA100
Embeddings	BGE-M3	568M	568M	8192	MIT	shared
Reranker	bge-reranker-v2	568M	568M	8192	MIT	shared
ASR	IndicConformer	600M	600M	streaming	MIT	shared
TTS	IndicParler-TTS	~1B	~1B	-	Apache 2.0	1xGPU
OCR	Surya OCR 2	~300M	~300M	-	GPL-3.0	shared
VLM	Qwen2.5-VL 7B	7B	7B	32K	Apache 2.0	1xGPU

API Endpoints

Endpoint	Method	Purpose
/api/chat	POST	Chat to RAG response
/ws/voice	WebSocket	Real-time voice I/O
/api/classify	POST	Formulation classification
/api/ocr	POST	Document image OCR
/api/escalate	POST	Request human facilitator
/api/audit	GET	Audit trail (admin)
/api/user/data	DELETE	DPDP: erase all data

The 5 Hackathon Differentiators

1. **Formulation Classifier works LIVE** — describes product, classifies, cites D&C Act
2. **Jurisdiction Toggle is REAL** — India + International answers, visually separated
3. **Citations are CLICKABLE** — [Section 3(p), Patents Act 1970] hyperlinked
4. **ABS Helper catches what people MISS** — "Under Biodiversity Amendment Act 2023, AYUSH practitioners using cultivated plants are exempt"
5. **Safe Abstention** — "This involves Section 3(e) analysis. Recommend IP attorney. [Escalate]"

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Research Sources: 10+ web sources, model benchmarks, API documentation